

EXPERIMENT 1 — Microsoft Threat Modeling

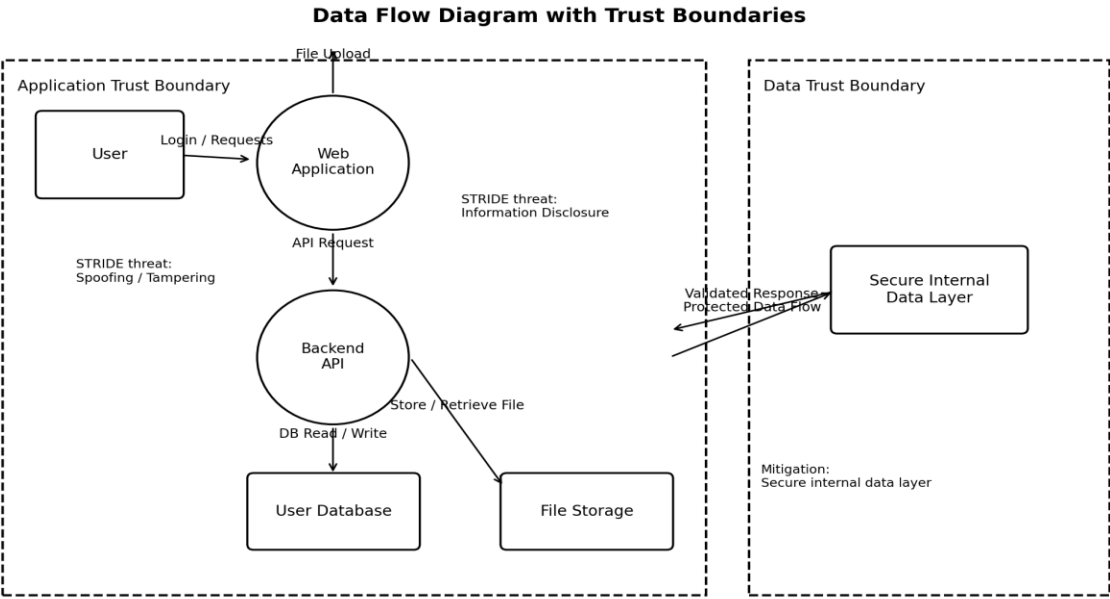
Aim

To identify potential security threats and vulnerabilities in a given application scenario using Microsoft Threat Modeling Tool, and to recommend appropriate security controls.

Procedure

1. Install and open Microsoft Threat Modeling Tool.
2. Create a new threat model.
3. Define the application scenario and its components.
4. Identify:
 - External entities
 - Processes
 - Data stores
 - Data flows
 - Trust boundaries
5. Draw the Data Flow Diagram (DFD) for the given scenario.
6. Define the properties of data flows and components.
7. Run the threat-generation/analysis feature.
8. Review the threats identified using the STRIDE methodology:
 - Spoofing
 - Tampering
 - Repudiation
 - Information Disclosure
 - Denial of Service
 - Elevation of Privilege
9. Analyze each identified threat and assign appropriate mitigation controls.
10. Generate/save the threat model report.

OUTPUT



Result

The given application scenario was successfully modeled, potential STRIDE-based threats were identified, and suitable security mitigations were recommended.